

CreditTrax

*Empowering Young Adults for Financial
Success*

Prototype Demo 1 Presentation
By Team: Copper
Old Dominion University
CS 411W
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Meet the Team



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Our Problem Statement

Many young adults in the U.S. struggle with their finances due to the overwhelming burden of student loans, ever increasing credit rates, and a lack of experience with crucial financial skills.

Balancing the pressures of starting a career, taking care of themselves and loved ones, all while facing financial uncertainty leads to the accumulation of debt, increased anxiety, and delayed life milestones such as marriage, home ownership, and starting a family.



The Problem Characteristics

Forgetting to pay monthly payments due to being busy

- ❖ Interest rate increase
- ❖ Takes even longer to pay it off
- ❖ Late fees
- ❖ Credit score sinks
 - Harder time making large purchases
- ❖ Mental and physical health decline
- ❖ Harder time achieving life goals.

Lack of proper budgeting skills and/or financial literacy

- ❖ Can't make realistic budgets
- ❖ Won't know where their money has gone
- ❖ Harder time prioritizing expenses, short-term and long-term goals
- ❖ Won't fully understand the impacts of late payments and general financial choices

Some apps have budget management features, some have debt management, not many have both

- ❖ May have to use at least two financial apps.
- ❖ Harder to manage finances
- ❖ Hard to see the bigger picture



Who is Affected?

Users

- ❖ **Young Professionals** overburdened by student loans and credit card debt.
- ❖ **Students and Recent Graduates** lacking financial literacy and seeking budgeting support.
- ❖ **Young Adults** (18-34) mentally overwhelmed by a lack of experience with finance.
- ❖ **Service members** transitioning from military to civilian struggling managing finances



Customers

- ❖ **Young Adults**
- ❖ **Financial Institutions** looking to partner with apps to better serve a younger audience.
- ❖ **Employers** looking to offer financial wellness programs for early-career employees.
- ❖ **Universities/Finance Professors** that want to offer an educational finance tool for students to learn personal finance



Stakeholders

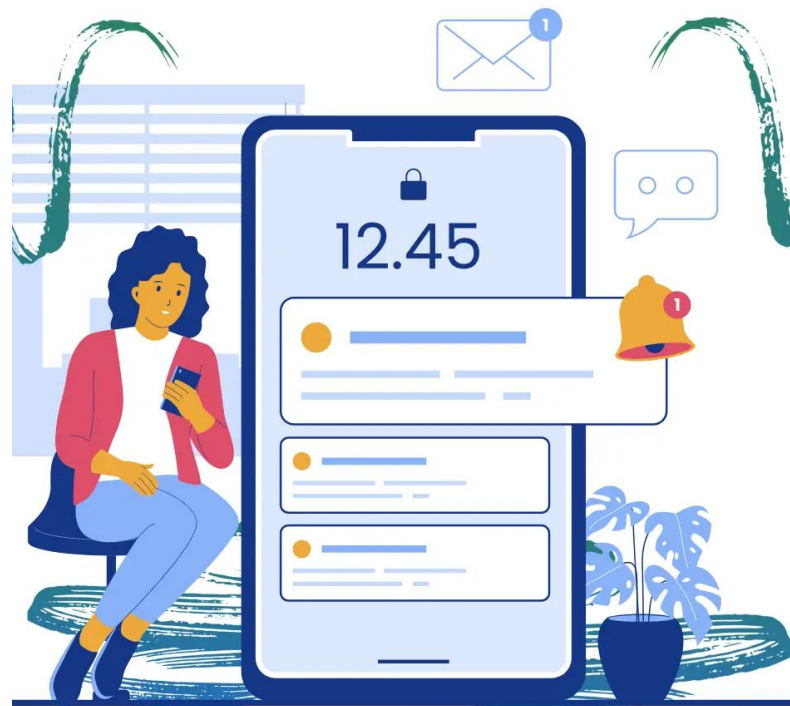
- ❖ **Financial Advisors** with an interest in helping young adults build sound financial habits
- ❖ **Financial Wellness Coaches** providing educational services to young adults to reduce debt and improve budgeting skills.
- ❖ **Credit Counselors** looking for tools to suggest to their younger clients.



Our Solution: CreditTrax

CreditTrax leverages AI-powered budgeting through personalized financial dashboards and interactive tools to help users align their spending with long-term financial goals while making smarter financial decisions.

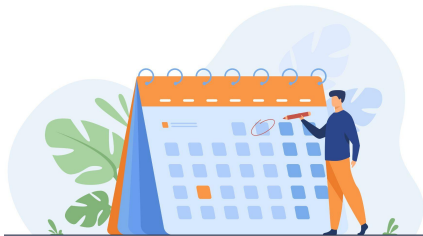
By analyzing spending habits, sending real-time notifications, and offering comprehensive financial literacy tools, CreditTrax enables young adults to achieve their financial objectives, track their progress, and reduce the stress and anxiety that often comes with managing debt.



Solution Characteristics

Debt Tracking

- ❖ Users can **input all their debts** e.g. credit card balances and student loans along with their respective interest rates and due dates
- ❖ **Customized Payment Reminders** prompts the user to **manually approve payment** of a debt. If they neglect to do so, the app will issue increasing reminders to ensure that they pay their debts on time.



Budgeting Recommendations

- ❖ Powered by **generative chatbot AI** to provide personalized recommendations based on a user's debt and spending habits
- ❖ Analyzes the user's income, expenses, and debt obligations to generate tailored suggestions to help them optimize their budget
- ❖ Reduces unnecessary spending and allocate funds more efficiently

For example:

If a user is just starting out in their career and is burdened with multiple loans, the app might suggest the **50-30-20** budget plan.

(50% for needs, 30% for wants, and 20% for savings and debt payments)

Personal Financial Dashboard

- ❖ **Data analytics** - tracks and analyzes users' spending habits and finances, allowing the dashboard to offer real-time visualizations and reports on financial health.
- ❖ **Visualization** - uses charts/graphs and badges to illustrate the user's progress with their debt. Encourages users to stay motivated to adhere to their budget and debt repayment plans.
- ❖ **"What-if" analysis** - This feature enables users to explore potential financial outcomes by simulating different scenarios, such as increasing loan payments or adjusting their savings, helping them make informed decisions based on projected results.

e.g. For users who manage to make every payment before the due date for 6 months straight.



Solution Characteristics

Financial Literacy

- ❖ **Builds essential financial skills:** CreditTrax educates users on key financial concepts like budgeting, saving, and managing debt, which are crucial for young adults who may lack experience in these areas.
- ❖ **Encourages proactive financial habits:** The app fosters good financial habits by teaching users how to balance debt payments with saving goals, enabling them to take control of their finances early in their careers.
- ❖ **Reduces financial anxiety:** CreditTrax helps young adults reduce financial stress by providing essential financial literacy skills, enabling them to effectively manage student loans and credit card debt.

Short vs Long-term Goals

- ❖ This app offers the user the flexibility of choosing between setting short-term or long-term goals to suit the user's changing needs.
- ❖ They can choose from a list of options or define their own goals, e.g. short term: building an emergency fund or eliminating debts and long term: buying a house or a car.



Sprint Breakdown

PROCESS	SPRINT 0		SPRINT 1		SPRINT 2		SPRINT 3		SPRINT 4		SPRINT 5		SPRINT 6	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Initial setup														
Set up Framework														
User Interface														
Feature Development														
System and Unit Testing														
Polish and Deployment														

Sprint Breakdown cont.

❖ Sprint 0 (Week 1-2)

- Setup development environment
- Create Trello board
- Create git repository

❖ Sprint 1 (Week 3-4)

- Set up frontend and backend frameworks
- Set up testing tools
- Setup APIs (Mostly OpenAI)
- Write initial unit tests
- Begin creating user stories

❖ Sprint 2 (Week 5-6)

- Develop basic UI functionality
 - Button presses
 - Page movement
- Begin developing core features:
 - What-if analysis, budgeting, debt tool
- Expand unit test

Sprint Breakdown cont.

❖ Sprint 3 (Week 7-8)

- Continue development of core features
- Begin development of other features
 - Dashboard, game, other users
- Finish creating unit tests
- Create/conduct performance tests
- Demo 1 in Week 7

❖ Sprint 4 (Week 9-10)

- Implement full functionality of features
- Create/conduct system tests

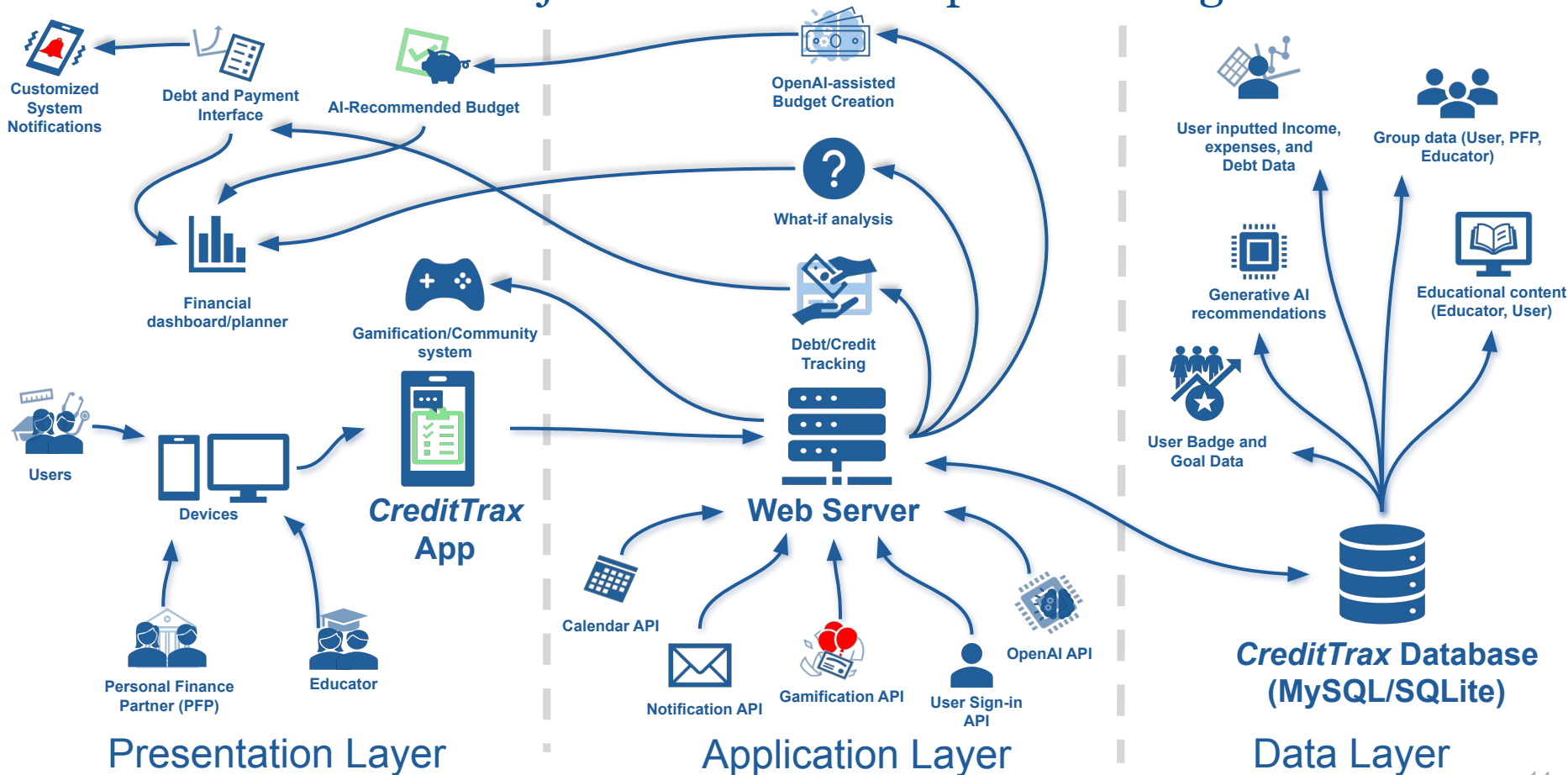
❖ Sprint 5 (Week 11-12)

- Finalize features
- Finalize UI functionality and form
- Review testing data
- Demo 2 in Week 11

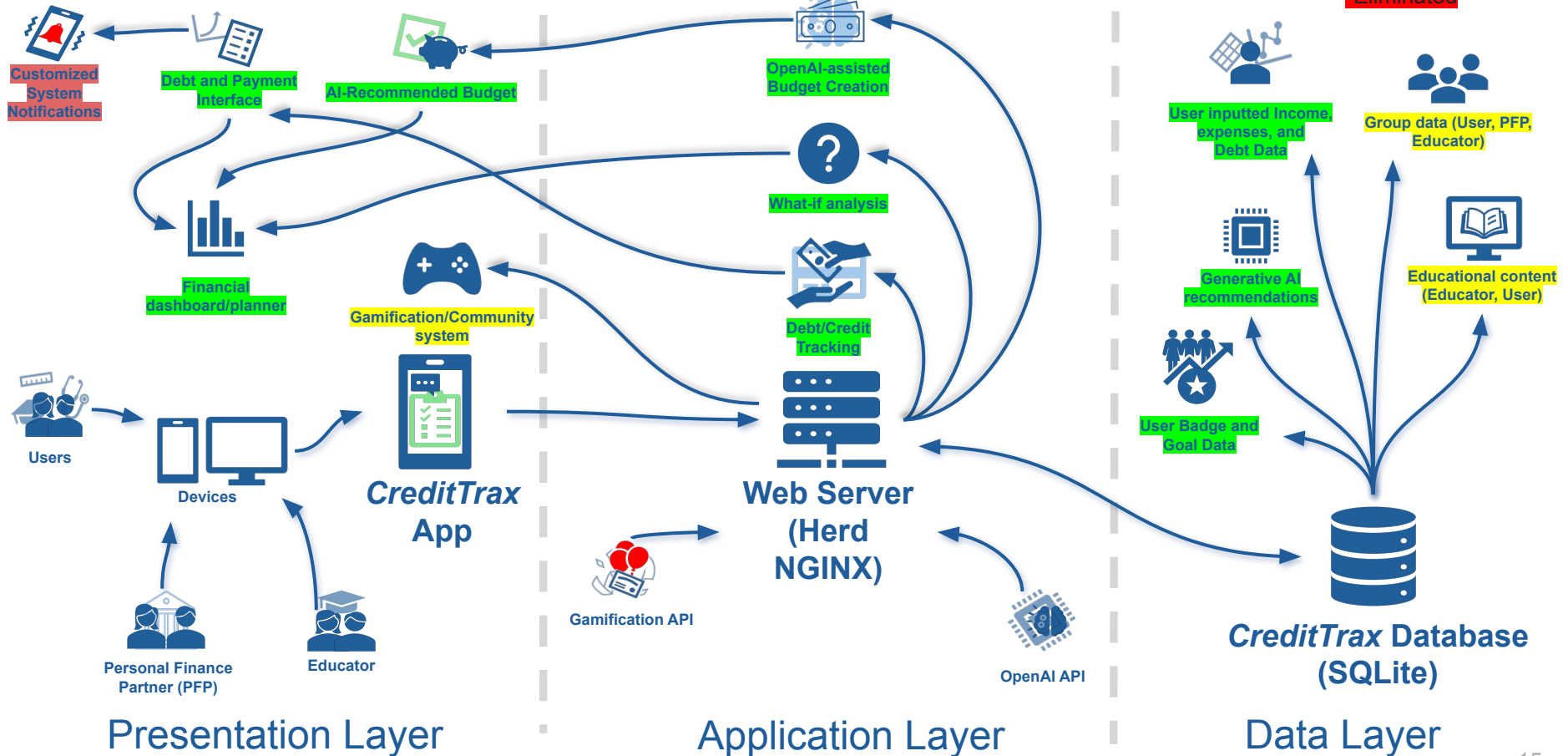
Sprint Breakdown cont.

- ❖ Sprint 6 (Week 13-14)
 - Create placeholder content for demo
 - Beta-test
 - Publish and present
 - Prototype Demo

RWP Major Functional Components Diagram



Prototype MFCD Diagram



Changes to scope, implementation language, etc.

- ❖ Shifted the focus from AI-assisted financial/debt management app to an AI-powered **educational tool** to teach financial literacy and give recommendations.
 - Still will implement the AI-assisted budgeting. Primary focus will be in what-if analysis
- ❖ Changed development tools and backend programming language
 - **Python → PHP**
 - **Added frameworks for HTML, CSS, JavaScript compatible with PHP**

Real World Product vs Prototype

*Fully implemented

*Partially implemented

*Eliminated

Functionality Group	Feature	RWP	Prototype	Notes
Account Management	Add/Delete Account	Full add/delete functionality for user accounts	Fully Implemented	Can use stand-in accounts
	Login/Authentication	MFA and session management	Fully Implemented	Use stand-in accounts
	Profile Management	Customizable user profile	Fully Implemented	Use stand-in accounts
Debt Tracking	Add/Edit Debt	Comprehensive tracking editing	Fully Implemented	
	Smart Payment Reminders	Customizable with escalation	Eliminated	Unsure of method used to send reminders to user without Google APIs
Budgeting	Personalized Budget Creation	AI-driven recommendations	Fully Implemented	Focus generative AI use in what-if analysis
	Expense Tracking	Detailed tracking with visuals	Fully Implemented	
	Budget Adjustments	AI-driven recommendations based on spending patterns	Fully Implemented	Focus generative AI use in what-if analysis

Real World Product vs Prototype Cont.

*Fully implemented
*Partially implemented
*Eliminated

Functionality Group	Feature	RWP	Prototype	Notes
Goal Tracking	Short/Long-Term Goals	Customizable short-term goal tracking	Fully Implemented	
	Goal Progress Visualization	Visual tracking	Fully Implemented	
What-If Analysis	AI-assisted projection/simulation	Use of AI for what-if	Fully Implemented	
	Create/Simulate scenarios	Algorithmic Impact simulations	Fully Implemented	
Analytics	Comprehensive Financial Dashboard	Budget and debt visualizations, goal progression,	Fully Implemented	
Reward System	Badges and Rewards	Custom badges for achievements	Partially Implemented	Basic badges for demonstration
	Community/Group page	Full community functionality with leaderboards	Partially Implemented	Can create groups
Financial Literacy Tools	Educational Resources	Upload external resources, show financial concepts embedded in other features	Partially Implemented	Only shows embedded finance information.

User Stories to Features

*Done
*WIP
*Planned

Functionality Group	Feature	User Story	User Role
Debt Tracking	Add, Edit, Delete and Track Debt	Create and edit a debt source with source name, amount due, interest rate, term length, and more	Young adults
		Delete a debt source	Young adults
		Create and edit a new debt transaction	Young adults
		Delete a debt transaction	Young adults
		See all my debt information condense into one easily accessible place	Young adults
	Smart Payment Reminders	Receive reminders for my payment due dates, so that I never miss a payment and avoid late fees.	Young adults
	Confirm payment of debt in app	Confirm payment of a debt once I paid the actual source in real life.	Young adults
AI-assisted Budgeting	Personalized Budget Creation	Use AI to help create a budget in a style based on my financial situation	Young adults
	Standard budget creation	Create a budget without the use of AI if I don't want to use it	Young adults

User Stories to Features cont.

*Done
*WIP
*Planned

Functionality Group	Feature	User Story	User Role
AI-assisted Budgeting	Create/edit/delete expense and category	Create, edit, and delete an expense transaction	Young adults
		Create, edit and delete an expense category	Young adults
		Create and edit a new income transaction	Young adults
		Delete an income transaction	Young adults
		Create, edit, and delete an income category	Young adults
	Track expenses	See the total amount of money I've spent on a category	Young adults
	AI-driven budget recommendations	Receive AI-driven budgeting suggestions, so that I can optimize my spending and savings based on my financial situation	Young adults
		See the latest recommendations from OpenAI model for my customer	Personal Finance Partner
	View the budget	View the budget that I created with or without the use of OpenAI model	Young adults
		View the budget that my customer created	Personal Finance Partner

User Stories to Features cont.

*Done
*WIP
*Planned

Functionality Group	Feature	User Story	User Role
Goal Tracking	Goal Progress Visualization	Track my short-term and long-term goals in one place	Young adults
		Automatically update my goals based on transactions logged into the app	Young adults
		Press a button confirming I have completed a goal	Young adults
		View all my completed goals	Young adults
		View and track my customer's goals	Personal Finance Partner
	Short-Term/ Long-term Goals	Have short-term goals so I can keep track of my personal finances in small milestones	Young adults
		Have long-term goals so I can set big financial milestones	Young adults
What-If Analysis	AI-assisted simulation/ recommendations	Use OpenAI model to ask for specific scenarios that the app won't be able to answer	Young adults
		Use OpenAI model to provide recommendations based on results of the algorithmic scenario.	Young adults

User Stories to Features cont.

*Done
*WIP
*Planned

Functionality Group	Feature	User Story	User Role
What-If Analysis	Scenario Predictions and Visualizations	See graphs that shows the amount of money I save in a period of time	Young adults
		See a graph that shows future debt payments over a period of time.	Young adults
	Algorithmic What-If Analysis	Be able to perform basic predictions on savings and debt without the use of OpenAI model	Young adults
		Have a way to run the what-if analysis using my customer's information	Personal Finance Partner
Financial Dashboard	Comprehensive summary of financial standing	See my debt to income ratio to help me not overextend myself	Young adults
		See my current total spending for this month	Young adults
		See current trends, pattern, and statistics generated by OpenAI model from analyzing my data.	Young adults
	Overview of goals	See a short-term goal, long-term goal, and progress for each of them	Young adults
	View achievements	See my last completed achievement	Young adults
		See an achievement that is the closest to completion	Young adults

User Stories to Features cont.

*Done
*WIP
*Planned

Functionality Group	Feature	User Story	User Role
Financial Dashboard	Access other pages from Dashboard	Go to the My Finances Page from the dashboard	Young adults
		Go to the Rewards page from the dashboard	Young adults
		Go to my Debt Table from the dashboard	Young adults
		Go to my goals section from the dashboard	Young adults
Gamification	Achievements	Earn achievements by completing certain requirements/milestones	Young adults
		Have the badges level up one tier when I earn it	Young adults

User Stories to Features cont.

*Done
*WIP
*Planned

Functionality Group	Feature	User Story	User Role
Community/Group	Groups	Partner with educators for advice on managing my debt	Young adults
		Join a group with my friends	Young adults
		Create a group	Young adults
		Way to add/delete users as my customers	Personal Finance Partner
		Add students to my personal class group	Educator
	Resources/Information	See accurate financial concepts throughout the app	Student

User Stories to Tasks

*Done
*WIP
*Planned

Functionality Group	Task	User Story	User Role
Debt Tracking	Create/Edit debt source	Create and edit a debt source with source name, amount due, interest rate, term length, and more	Young adults
	Delete debt source	Delete a debt source	Young adults
	Create/edit debt transaction	Create and edit a new debt transaction	Young adults
	Delete debt transactions	Delete a debt transaction	Young adults
	View debt sources and transactions	See all my debt information condense into one easily accessible place	Young adults
	Create/edit/delete/see/receive reminders	Receive reminders for my payment due dates, so that I never miss a payment and avoid late fees.	Young adults
	Create pop up screen asking for confirmation	Confirm payment of a debt once I paid the actual source in real life.	Young adults
AI-assisted Budgeting	Integrate gpt-4o-mini to budget creation	Use AI to help create a budget in a style based on my financial situation	Young adults
	Include option to use standard budget during creation process	Create a budget without the use of AI if I don't want to use it	Young adults

User Stories to Tasks cont.

*Done
*WIP
*Planned

Functionality Group	Task	User Story	User Role
AI-assisted Budgeting	CRUD a transaction	Create, edit, and delete an expense/income transaction	Young adults
	CRUD a category	Create, edit and delete an expense/income category	Young adults
	View expenses/income	See the total amount of money I've spent on a category	Young adults
	AI-driven budget recommendations	Receive AI-driven budgeting suggestions, so that I can optimize my spending and savings based on my financial situation	Young adults
		See the latest recommendations from OpenAI model for my customer	Personal Finance Partner
	View the budget	View the budget that I created with or without the use of OpenAI model	Young adults
		View the budget that my customer created	Personal Finance Partner

User Stories to Tasks cont.

*Done
*WIP
*Planned

Functionality Group	Task	User Story	User Role
Goal Tracking	View goals	Track my short-term and long-term goals in one place	Young adults
	Update goals automatically	Automatically update my goals based on transactions logged into the app	Young adults
	Manual confirmation of goal completion	Press a button confirming I have completed a goal	Young adults
	Record and view past goals	View all my completed goals	Young adults
	View customer's goals	View and track my customer's goals	Personal Finance Partner
	CRUD Short-Term/ Long-term Goals	Have short-term goals so I can keep track of my personal finances in small milestones	Young adults
		Have long-term goals so I can set big financial milestones	Young adults
What-If Analysis	Integrate gpt-4o-mini chat box	Use OpenAI model to ask for specific scenarios that the app won't be able to answer	Young adults
	Have gpt-4o-mini read results of what-if analysis, generate response	Use OpenAI model to provide recommendations based on results of the algorithmic scenario.	Young adults

User Stories to Tasks cont.

*Done
*WIP
*Planned

Functionality Group	Task	User Story	User Role
What-If Analysis	Show graphs of results	See graphs that shows the amount of money I save in a period of time	Young adults
		See a graph that shows future debt payments over a period of time.	Young adults
	Implement algorithmic what-if analysis with calculations	Be able to perform basic predictions on savings and debt without the use of OpenAI model	Young adults
		Have a way to run the what-if analysis using my customer's information	Personal Finance Partner
Financial Dashboard	View updated financial information	See my debt to income ratio to help me not overextend myself	Young adults
		See my current total spending for this month	Young adults
	Generate trends based on user data with gpt-4o-mini	See current trends, pattern, and statistics generated by OpenAI model from analyzing my data.	Young adults
	View goals	See a short-term goal, long-term goal, and progress for each of them	Young adults
	View achievements	See my last completed achievement	Young adults
		See an achievement that is the closest to completion	Young adults

User Stories to Tasks cont.

*Done
*WIP
*Planned

Functionality Group	Task	User Story	User Role
Financial Dashboard	Access other pages from Dashboard	Go to the My Finances Page from the dashboard	Young adults
		Go to the Rewards page from the dashboard	Young adults
		Go to my Debt Table from the dashboard	Young adults
		Go to my goals section from the dashboard	Young adults
Gamification	Update achievement progress	Earn achievements by completing certain requirements/milestones	Young adults
	Design badge tiers	Have the badges level up one tier when I earn it	Young adults

User Stories to Tasks cont.

*Done
*WIP
*Planned

Functionality Group	Task	User Story	User Role
Community/Group	Link accounts to a group	Partner with educators for advice on managing my debt	Young adults
		Join a group with my friends	Young adults
	CRUD a group	Create a group	Young adults
		Way to add/delete users as my customers	Personal Finance Partner
		Add students to my personal class group	Educator
	Embed basic financial concepts to app	See accurate financial concepts throughout the app	Student

Development Tools

Development Type	Front-End	Back-end
IDE	VSCode	VSCode
Version Control	Github	Github
CI and CD	Github	Github
Language	HTML5, CSS, JS	PHP
Testing Framework	Jest	PHPUnit
Documentation Tool	JSDoc	Scribe

Required Libraries, Tools, & Technologies

❖ Libraries

- MathPHP
- Chart.js

❖ Tools

- Herd: Local Development Environment Manager
- Laravel: Backend Framework for PHP
- TailWind: Frontend Framework for CSS
- Alpine: Frontend Framework for JS
- Composer: Dependency manager
- SQLite: Database Management System

❖ Technologies (Hardware and Software)

- OpenAI API
- GameLayer API

Github Repository

- Everyone in Team Copper, Mr. Jacob Cook, and Professor Hosni should have access to the repository.
- Github Repository: <https://github.com/bnguyen117/s25-Team-Copper>

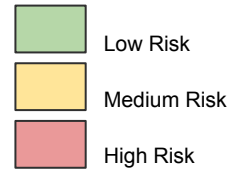
User Risks

User Risks

- **UR-1:** The tool's UI is too complicated, making data entry, tracking, and real-time management difficult and prone to errors.
- **UR-2:** User forgot to respond to CreditTrax when the monthly bill was paid.
- **UR-3:** User noticed that a feature is buggy or partially functioning.
- **UR-4:** User's app crashed
- **UR-5:** User finds their current budget style unsatisfactory.

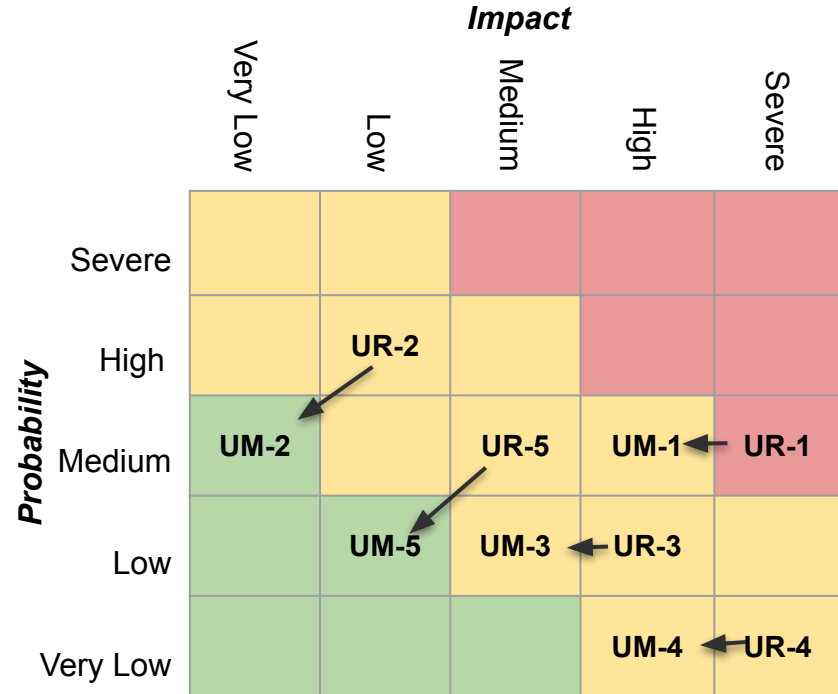
*Done
*WIP
*Planned

Legend



Risk Mitigation

- **UM-1:** Simplify the UI to enhance user experience, making data entry and tracking more intuitive and reducing the likelihood of errors. Provide tutorials or guided walkthroughs to help users navigate the app effectively.
- **UM-2:** Asks users if they paid the bill the next time they open the app.
- **UM-3:** User can send a bug report and developers will patch the bug.
- **UM-4:** User will be notified for reason of crash and can send a crash report to developers.
- **UM-5:** CreditTrax can advise user another budgeting technique (50/30/20, zero-based, etc.)



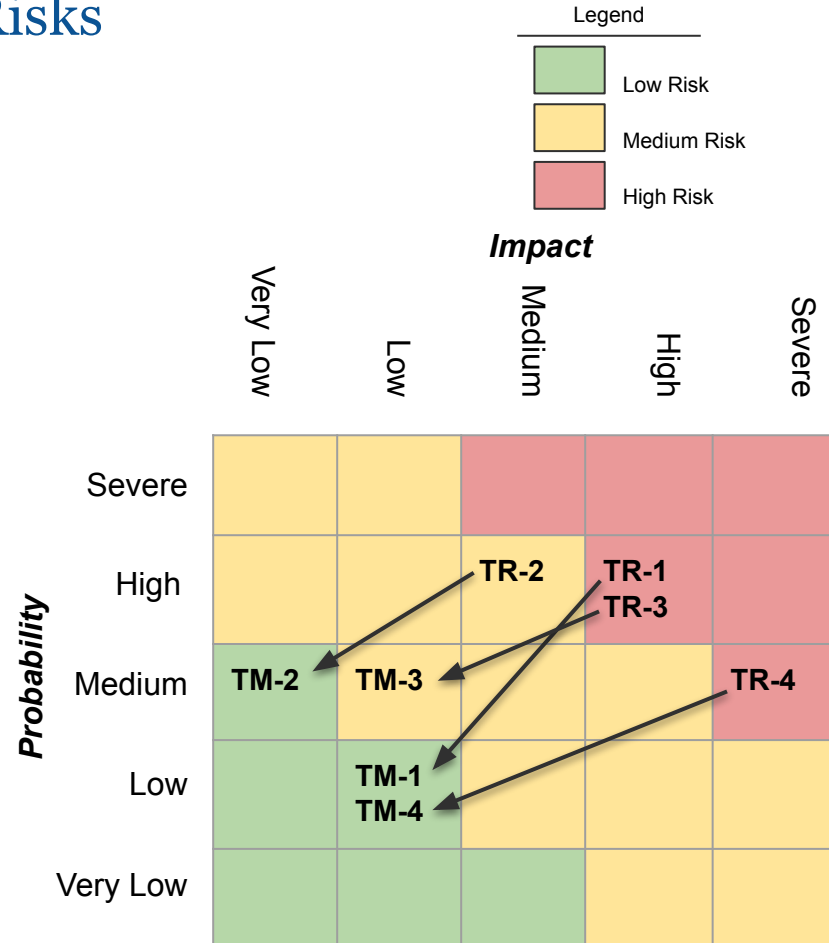
Technical Risks

Technical Risks

- **TR-1:** Integration of AI-driven budgeting recommendations and multiple features that can be complex
- **TR-2:** Ensure the app runs smoothly across different smartphones, operating systems, and versions.
- **TR-3:** Issues with handling concurrent users or processes at one time that also may make the application struggle to scale effectively.
- **TR-4:** The developed solution may not fully meet users' actual needs, leading to potential disengagement with the app.

Risk Mitigation

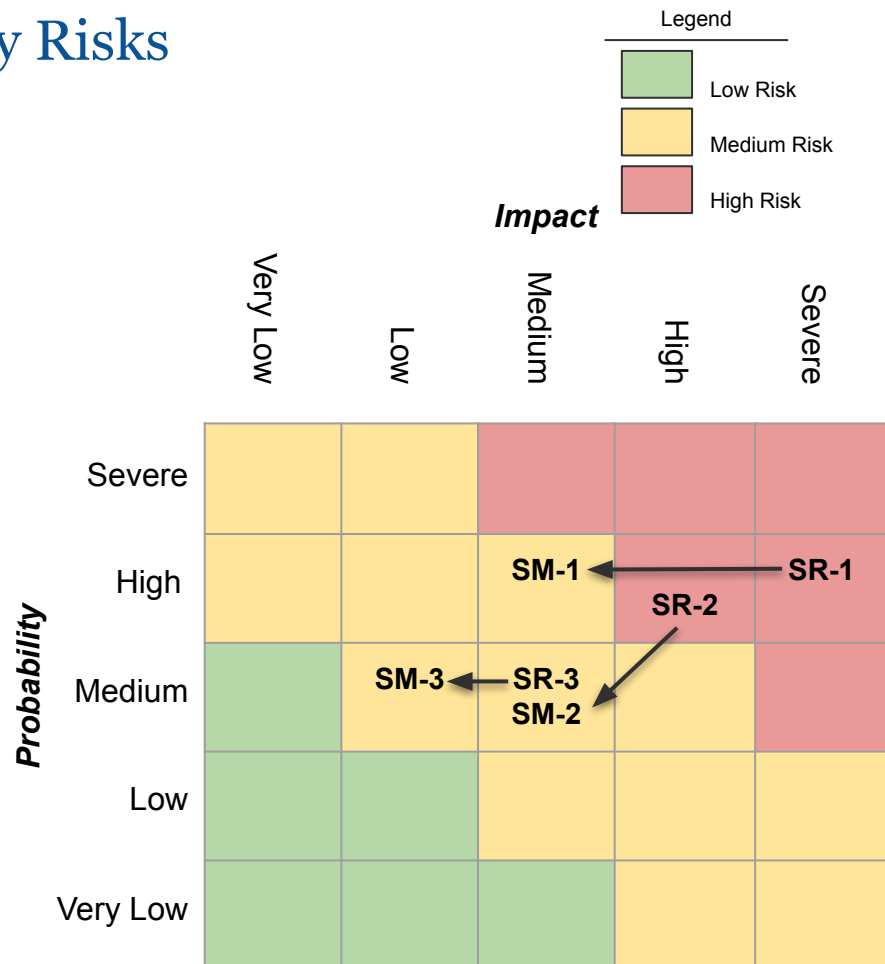
- **TM-1:** Break down the development process into smaller, manageable tasks; using modular architecture and thorough testing.
- **TM-2:** Develop a robust compatibility testing plan that covers a wide range of devices and OS versions.
- **TM-3:** Implement concurrency control mechanisms and perform load testing with the use of scalable infrastructure and cloud services
- **TM-4:** Engage with end-users throughout the development process through surveys and interviews, and implement regular updates and feedback mechanisms to keep users engaged.



Security Risks

- ❖ Security Risks
 - **SR-1:** Risks related to data breaches, leading to unauthorized access to user's sensitive financial information.
 - **SR-2:** Weak encryption mechanisms might expose sensitive data during storage or transmission.
 - **SR-3:** Inadequate authentication and authorization protocols may allow unauthorized users to access.

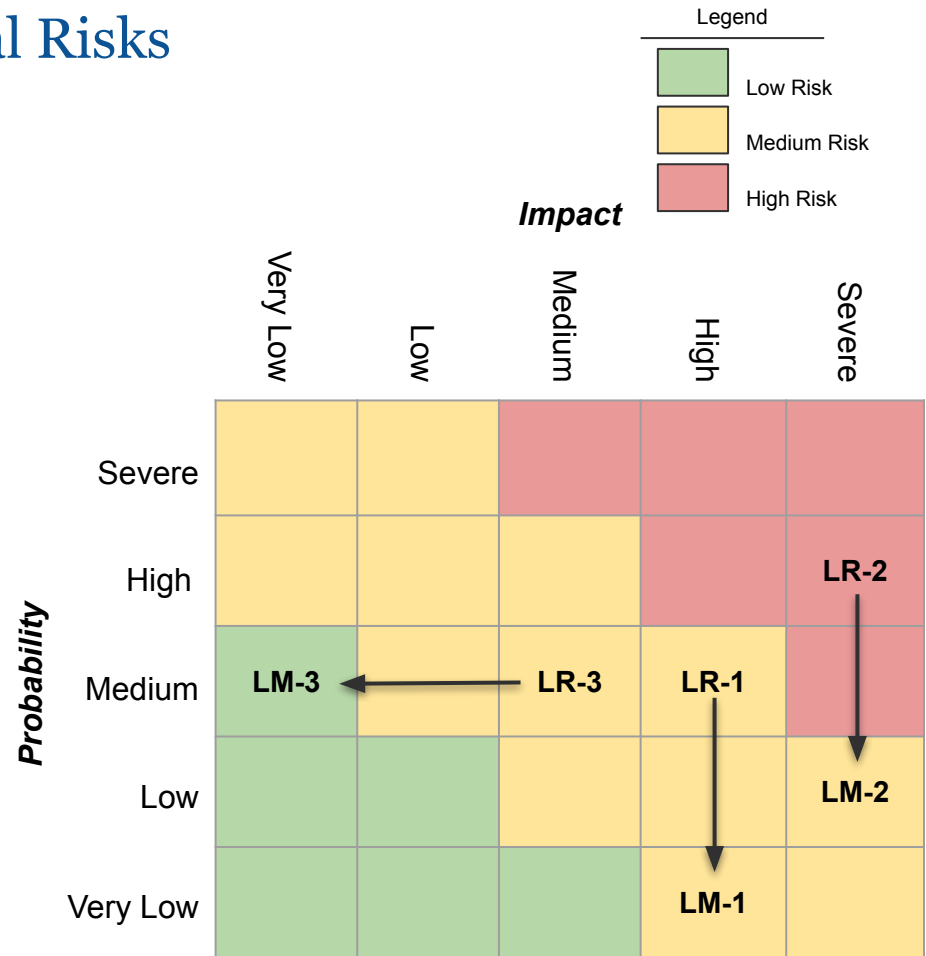
- ❖ Risk Mitigation
 - **SM-1:** Implement robust security practices such as data encryption (both at rest and in transit), multi-factor authentication (MFA), and regular security audits.
 - **SM-2:** Use AES-256 provided by Amazon RDS with real-time monitoring and regular audits to thwart breaches.:
 - **SM-3:** Keep systems updated, conduct ongoing security training, and have a rapid incident response plan to quickly tackle threats.



Legal Risks

- ❖ Legal Risks
 - **LR-1:** Copyright infringement on competitors
 - **LR-2:** User data privacy violated by data breaches
 - **LR-3:** AI-generated recommendations harming users' financial health

- ❖ Risk Mitigation
 - **LM-1:** Extensive research on competitors' features, design, and trademarks to avoid infringement
 - **LM-2:** Implementation of security software and encryption of user data stored in database
 - **LM-3:** Statements shown on account creation page informing users of AI usage and that investment recommendations don't have a 100% guarantee of success



Issues and Challenges

- Learning curve
 - Mostly getting used to learning by doing.
- GameLayer API may require an account with a work email address to use the API. Not confirmed yet.

Next Steps: Brian

- Research further into OpenAI API capabilities in web apps.
 - Experimenting the API in CreditTrax
 - Explore other options such as machine learning algorithms, Claude, etc.
- Begin implementation of OpenAI API into What-if analysis and finances
 - Implement chatbox to what-if analysis
 - Implement background analysis of what-if analysis report
 - Implement background analysis of user finance data and generate trends
- Learn how to integrate GameLayer API, or find another suitable gamification API for Laravel
- Help develop the budget creation process
- Begin developing functionality of budget.
- Revamp database schema based on currently implemented database and plan further.

Next Steps: Bryce

- Fine-tune the Dashboard page for improved usability
 - Refine the design of graphs for better visualization
 - Implement routes to specific destinations for seamless navigation
 - Modify charts to capture and display real-time user input
- Research and integrate new features to enhance app functionality
- Optimize performance for a faster and smoother user experience
- Ensure UI consistency across all dashboard elements for a cohesive design

Next Steps: Darrick

- Start creating What-If analysis algorithms that can interact with the user debt, financial goal, and budgeting data in our database.
- Link these algorithms as options in the What-If analysis form.
- Create a new page where users will be sent to view their What-If analysis report after form submission.
- Start learning Filament Charts, so we can visually represent the data within these reports to the user.
- Store these reports in a database table so users can revisit

Next Steps: Erika

- Refine Rewards / Achievements page by:
 - Integrating GameLayer API to the app so we can have dynamic badges (if we go through with it)
 - Making the UI functionalities work with the backend
 - Remove Goals section from Rewards page since it's already in the Finance page
 - OR make 'Add Goal' button link to Finance page instead but still show the goals the user already has
- Learn more HTML and CSS

Next Steps: Indi

- Finish the community page
- So far the friends can request and accept and delete each other
- Next is the groups, with updating groups and display points from the rewards

Next Steps: Josh

- Add method of creating transactions to add new sources of income or debt to a user's budget.
- Add a hard-coded budgeting algorithm based off of the 50/30/20 model of budgeting.
- Add a viewable budgeting breakdown to Finance page.

Next Steps: Smit

- Retrieve and Display Budget Data:
 - Fetch and display user-specific budget information from the database on the Budget UI.
 - Set up forms and backend logic to allow users to update and save their budget data.
 - Make the budget page responsive and improve user interaction with visual feedback and cues.



Q & A

We welcome your
questions!

Appendix

...

User Stories

As a *Young Adult*, I would like to...

Financial Dashboard:

- Track my financial goals using a personalized dashboard so that I can stay motivated and reduce financial stress.

What-if analysis:

- Run a "what-if" analysis on my debt repayment strategies so that I can understand the long-term impact of different financial decisions.
- Use an algorithmic what-if analysis for simple financial scenarios.
- Use AI to analyze and generate recommendations based on results of an algorithmic what-if analysis.

Debt tracking

- Create a debt source with source name, amount due, interest rate, term length, and more
- Edit a debt source with source name, amount due, interest rate, term length, and more
- Delete a debt source
- Receive reminders for my payment due dates, so that I never miss a payment and avoid late fees.

User Stories

As a *Young Adult*, I would like to...

Debt-tracking cont:

- Add, edit, and delete new debt transaction
- See all my debt information condensed into one easily-accessible place.
- Confirm payment of a debt in the app once I paid the actual source in real life.

AI-assisted Budgeting:

- Receive AI-driven budgeting suggestions, so that I can optimize my spending and savings based on my financial situation.
- Create a budget without the use of AI if I don't want to use it.
- Use AI to help create a budget in a style based on my financial situation

Community/Groups

- Create a group with friends.
- Join a community/group with other people, my educator, or my personal finance partner.

User Stories

As a *Young Adult*, I would like to...

Gamification

- Earn achievements for handling my finances.
- Stay on track of my finances through goals.

User Stories

As a *Personal Financial Partner*, I would like to...

- See my clients' progress on their debt and repayments so that I can provide better guidance and financial solutions.
- Access and review AI budget recommendations provided to my clients.
- Customize budget recommendations based on my client's unique financial circumstances.
- Provide incentives for my client to motivate them toward developing good financial habits.
- Create groups with my clients.
- See my client's financial dashboard to get a general view of their financial health.
- Provide accessible resources to my client that teaches them how to manage their finances.

User Stories

As an ***Educator***, I would like to...

- Find and enroll users as my students.
- Set financial goals for my students to work toward.
- View and track my students' progress toward achieving their financial goals.
- View my students' achievement lists to track their progress in their financial journey without invading their privacy.
- Share helpful resources with my students to help them with their financial education.

User Stories

As a **Tester**, I would like to...

- As a tester, I would like to verify that the personal financial dashboard correctly visualizes user spending and debt progress so that the data is accurately represented.
- As a tester, I would like to test the Ai-powered recommendations to ensure that they are relevant and helpful for different financial situations.
- As a tester, I would like to validate that the “what-if” analysis accurately calculates projected financial outcomes based on different inputs.
- As a tester, I would like to simulate various user behaviors to ensure that the app provides appropriate alerts and reminders.